

Digital Electronics –

1. Number System

- .
- Binary
0 1 .
- Hexadecimal memory address programming .
- Octal permissions .

System	Base	Digits	
Binary	2	0,1	
Octal	8	0 - 7	Legacy systems
Decimal	10	0 - 9	
Hexadecimal	16	0 - 9,A - F	Programming

2. Boolean Algebra

- Boolean Algebra value : 0 1.
- Digital logic design .
- Boolean laws circuits simplify .
- De Morgan theorem NAND/NOR implementation .

Law	Equation	
Identity	$A + 0 = A$	
Complement	$A + A' = 1$	value
Idempotent	$A + A = A$	Same variable
Absorption	$A + AB = A$	

3. Logic Gates

- Logic gates building blocks .

- AND gate logical multiplication .
- OR gate logical addition .
- NOT gate input invert .
- XOR gate equality check .

Gate	Equation	
AND	$Y = A \cdot B$	Condition checking
OR	$Y = A + B$	Alternative logic
NOT	$Y = A'$	Inversion
XOR	$A \oplus B$	Adder circuits
XNOR	Equality	Comparator

4. K - Map Simplification

- Karnaugh Map Boolean expression simplify .
- circuit gates, power cost.
- Grouping power of 2 .

Group	Variables Remove
2	1
4	2
8	3

5. Adders and Subtractors

- Half adder bits add .
- Full adder bits add .
- Carry look - ahead adder fast .
- Subtractor binary subtraction .

Circuit	Inputs	Outputs	
Half Adder	2	2	Simple addition
Full Adder	3	2	Complex addition
Half Subtractor	2	2	Binary subtraction
Full Subtractor	3	2	Advanced subtraction

6. Flip - Flops

- Flip - flop bit memory store .
- Sequential circuits memory element .
- D flip - flop registers .
- JK flip - flop universal flip - flop .

Flip - Flop	Feature	
SR	Set/Reset	Basic storage
D	Data store	Registers
T	Toggle	Counters
JK	Universal	General purpose

7. Counters

- Counter clock pulses .
- Ripple counter slow propagation delay .
- Synchronous counter fast .

Counter	Feature	
Ripple	Asynchronous	Simple counting
Synchronous	Fast	High speed systems
Ring	One - hot	Sequence generation
Johnson	Twisted ring	Timing circuits

8. Registers

- Register flip - flops group .
- store transfer .
- Shift register data left/right shift .

Type	Meaning	
SISO	Serial In Serial Out	Serial transfer
SIPO	Serial In Parallel Out	Data conversion

Type	Meaning	
PISO	Parallel In Serial Out	Communication
PIPO	Parallel In Parallel Out	Storage

9. Decoder, Encoder, Multiplexer

- Multiplexer inputs select .
- Decoder binary input unique output convert .
- Encoder inputs binary code convert .

Device	Function	
MUX	Data selector	Routing
DEMUX	Data distributor	Distribution
Decoder	Binary to output	Memory selection
Encoder	Input coding	Keyboard encoding

Shortcut Notes

- XOR → Different input = 1
- XNOR → Same input = 1
- Left Shift → $\times 2$
- Right Shift → $\div 2$
- NAND NOR → Universal Gates
- D Flip - Flop → Data storage
- JK Flip - Flop → Universal Flip - Flop
- Synchronous Counter → Fast
- Ripple Counter → Slow